

FINAL PROJECT – WEB DEVELOPMENT

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PROJECT IDEA

The project is a web application that helps snowboarders and skiers plan and manage trips by recommending suitable ski resorts based on weather conditions, user preferences, and skill level.

Users create an account and define their preferences (e.g., ski/snowboard, skill level, terrain type). Based on selected travel dates and real-time snow forecast data, the system recommends the best resorts and provides personalized insights.

In addition to trip planning, the application also supports in-resort usage, allowing users to explore resort maps and receive recommendations for areas, facilities, and activities while at the resort.

The system is designed for winter sports enthusiasts who want a smarter and more personalized way to plan and experience their trips.

PROJECT PURPOSE

Planning a ski or snowboard trip requires checking multiple sources such as weather forecasts, snow reports, and resort information. This process is time-consuming and requires experience to interpret correctly.

This project simplifies the process by:

- ⇒ centralizing weather and resort data
- ⇒ providing personalized recommendations using AI
- ⇒ saving user preferences for repeated use
- ⇒ allowing users to plan and manage multiple trips
- ⇒ supporting users both before and during the trip

The system improves decision-making and enhances the overall trip experience.

MAIN COMPONENTS

FRONTEND:

The frontend handles user interaction and displays all information.

Main responsibilities:

- ⇒ user registration and login
- ⇒ collecting user preferences
- ⇒ displaying recommended resorts
- ⇒ showing weather and snow conditions
- ⇒ presenting AI-generated summaries
- ⇒ displaying resort maps and in-resort recommendations
- ⇒ managing saved trips and user profile

BACKEND:

The backend manages logic, data processing, and API communication.

Main responsibilities:

- ⇒ authentication and user management
- ⇒ storing and retrieving user preferences
- ⇒ fetching weather and snow data from external APIs
- ⇒ calculating resort suitability scores
- ⇒ managing trips and saved data
- ⇒ communicating with the AI component
- ⇒ handling in-resort search and recommendations

DATABASE (MySQL):

The database stores persistent data.

Main entities:

- ⇒ users
- ⇒ user preferences
- ⇒ resorts
- ⇒ weather logs
- ⇒ trips
- ⇒ resort locations (points of interest inside resorts)

AI COMPONENTS:

1. Resort Suitability Summary

Generates a personalized explanation of whether a selected resort is a good choice based on:

- ⇒ snow forecast
- ⇒ season timing
- ⇒ user skill level
- ⇒ terrain characteristics

2. Resort Comparison & Recommendation

Analyzes multiple resorts and returns the top 3 options based on:

- ⇒ snowfall and snow depth
- ⇒ weather conditions
- ⇒ terrain suitability
- ⇒ user preferences

3. Gear Recommendation Assistant

Provides gear suggestions based on:

- ⇒ forecasted weather conditions
- ⇒ snow type
- ⇒ trip duration
- ⇒ rider level

4. *In-Resort Recommendation Assistant*

While at the resort, the AI helps users:

- ⇒ find suitable slopes based on skill level
- ⇒ locate facilities (restaurants, rentals, lifts)
- ⇒ explore terrain areas (beginner zones, parks, advanced areas)

BASIC API DESIGN

Authentication:

- ⇒ **POST /auth/register**
Registers a new user
Request: name, email, password, preferences
Response: success message + user data
- ⇒ **POST /auth/login**
Logs in a user
Request: email, password
Response: authentication token

User & Preferences:

- ⇒ **GET /profile**
Returns user profile and preferences
- ⇒ **PUT /preferences**
Updates user preferences

Resorts:

- ⇒ **GET /resorts**
Returns list of resorts
- ⇒ **GET /resort/:id/forecast**
Returns weather and snow data for a resort

Recommendations:

- ⇒ **POST /recommend-resorts**
Returns top 3 resort recommendations
Request: dates (preferences optional if logged in)
Response: resorts + scores + AI explanation
- ⇒ **POST /resort-summary**
Returns AI summary for selected resort
Request: resort ID, dates
Response: suitability summary

Trips:

- ⇒ **POST /trips**
Creates a new trip
- ⇒ **GET /trips**
Returns user's saved trips

Gear Assistant

- ⇒ **POST /gear-recommendation**
Returns gear suggestions

In-Resort Features:

- ⇒ **GET /resort/:id/map**
Returns resort map and locations
- ⇒ **GET /resort/:id/locations**
Returns points of interest (lifts, restaurants, etc.)
- ⇒ **POST /resort-assistant**
Returns AI recommendations inside the resort

PAGES/SCREENS

I. Home Page

- ⇒ main landing page of the application
- ⇒ includes:
 - search bar for resorts
 - date selection (trip dates)
 - optional filters (region, country)
- ⇒ displays list of resorts or search results
- ⇒ navigation bar includes sign in / sign up option
- ⇒ allows users to browse without logging in

II. Top 3 Recommendations Page

- ⇒ user selects:
 - trip dates
 - (if not logged in) skill level and terrain preferences
- ⇒ system displays the top 3 recommended resorts in a **podium or ranked layout**
- ⇒ each resort card includes:
 - name
 - short explanation
 - key weather/snow data
- ⇒ clicking a resort redirects to the Resort Details Page

III. Resort Details Page

- ⇒ displays detailed information about a selected resort:
 - weather forecast and snow conditions
 - terrain and difficulty level
 - AI-generated suitability summary
- ⇒ includes option to:
 - save the trip (for logged-in users)
- ⇒ includes access to:
 - resort map
 - in-resort recommendations

IV. Resort Explorer Page (In-Resort Use)

- ⇒ designed for use during the trip
- ⇒ displays:
 - resort map (image or embedded map)
 - key locations (lifts, restaurants, rental shops, slopes)
- ⇒ allows users to:

- search for locations within the resort
- view categorized points of interest
- ⇒ includes AI-based recommendations such as:
 - suitable areas based on skill level
 - suggested zones (beginner areas, parks, advanced terrain)
 - nearby facilities

V. Login / Signup Page

- ⇒ allows users to create an account or log in
- ⇒ during registration, users define preferences:
 - ski or snowboard
 - skill level
 - terrain preference
- ⇒ preferences are saved for future use

VI. Saved Trips Page

- ⇒ available only for logged-in users
- ⇒ displays a list of saved trips
- ⇒ each trip includes:
 - resort
 - dates
- ⇒ allows users to revisit and manage their planned trips

VII. Profile / Preferences Page

- ⇒ allows users to view and update personal preferences
- ⇒ preferences include:
 - sport type
 - skill level
 - terrain preference

DATA

Users

- ⇒ id
- ⇒ name
- ⇒ email
- ⇒ password

User Preferences

- ⇒ user_id
- ⇒ sport_type (ski/snowboard)
- ⇒ skill_level
- ⇒ terrain_preference

Resorts

- ⇒ id
- ⇒ name
- ⇒ country
- ⇒ elevation
- ⇒ terrain_type
- ⇒ difficulty_level

- ⇒ latitude
- ⇒ longitude

Weather Logs

- ⇒ id
- ⇒ resort_id
- ⇒ date
- ⇒ snowfall
- ⇒ snow_depth
- ⇒ temperature
- ⇒ wind_speed

Trips

- ⇒ id
- ⇒ user_id
- ⇒ resort_id
- ⇒ start_date
- ⇒ end_date

ResortLocations

- ⇒ id
- ⇒ resort_id
- ⇒ name
- ⇒ type (lift, restaurant, slope, rental, etc.)
- ⇒ description

AI INTEGRATION IDEAS

- ⇒ generate personalized summaries of resort suitability based on forecast data and user preferences
- ⇒ compare multiple resorts and explain top recommendations
- ⇒ suggest gear based on weather and snow conditions
- ⇒ provide in-resort recommendations for slopes, facilities, and areas based on rider level

BASIC WORKFLOW

1. User enters the website and browses resorts or searches by name and dates
2. User can optionally sign up and define preferences
3. User navigates to the Top 3 Recommendations page and enters trip dates
4. Backend retrieves weather data and calculates scores
5. System displays top 3 resorts with AI explanations
6. User selects a resort and views detailed information
7. User can save the trip (if logged in)
8. User receives gear recommendations if needed
9. During the trip, user accesses the Resort Explorer:
 - 9.1. views map
 - 9.2. searches locations
 - 9.3. receives AI suggestions